

A STUDY ON ACTIVATION FUNCTIONS IN ARTIFICIAL NEURAL NETWORKS FOR HEAVY RAINFALL PREDICTION IN HO CHI MINH CITY

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Abstract

This study evaluates the performance of seven activation functions in Time-Delay Neural Networks (TDNN) for forecasting monthly maximum rainfall in Ho Chi Minh City using data from Tan Son Hoa station (1980–2024). Two TDNN architectures, (12-10-10-1) and (12-15-15-1), were tested with ReLU, Sigmoid, PReLU, RReLU, ELU, LeakyReLU, and GELU. The configuration (12-15-15-1) with RReLU yielded the best performance, achieving the lowest RMSE and highest R^2 . These results highlight RReLU as the most effective activation function for rainfall prediction in the study area and reinforce the potential of TDNN models in hydrometeorological forecasting.

Keywords: Activation Function, ANN, TDNN, Extreme Rainfall.

1. INTRODUCTION

Rainfall forecasting, particularly extreme precipitation events, remains one of the major challenges in hydrometeorology. Monthly maximum rainfall is critical both for scientific understanding and for practical applications such as disaster risk management, urban planning, and flood prevention, especially in large metropolitan areas like Ho Chi Minh City. In the context of global climate change and the increasing frequency of extreme weather events, the development of forecasting methods with high accuracy and strong stability is urgently needed.

In recent years, machine learning and deep learning have shown clear advantages over traditional statistical methods in hydrometeorological forecasting. Studies such as [1] and Rasp & Lerch (2018) [2] demonstrated that models like ConvLSTM and deep neural networks can outperform ARIMA and ensemble systems by better capturing spatiotemporal features. In Vietnam, Phu Nguyen-Duc et al. (2024) [3] confirmed that LSTM models achieved low errors ($MAE/RMSE \approx 0.2-0.3$) and high correlation ($R^2 = 0.8-0.9$) in monthly rainfall prediction.

Among neural architectures, TDNN is well-suited for time series with delayed and nonlinear dependencies. Initially successful in speech recognition [4], TDNN has since been applied to hydrological forecasting [5]. Recent work [6] in India showed that TDNN enhanced with wavelet-based activation functions improved rainfall prediction accuracy and reduced agricultural risks.

Activation function choice is a key factor affecting TDNN performance. Traditional functions like Sigmoid and Tanh suffer from gradient saturation, while modern functions (ReLU, Leaky ReLU, ELU, RReLU) improve convergence and generalization [7–8]. In rainfall estimation, Sattari et al. (2023) [9] found ExpRectifier to yield the best RMSE, while Øydvin et al. (2024) [10] reported that ReLU performed better in deeper networks but was more prone to overfitting compared to logistic activation, which provided more stable generalization.

Rainfall in Ho Chi Minh City is highly skewed, strongly seasonal, and influenced by monsoon dynamics, ENSO, and local convection. Extreme rainfall events, often exceeding 300–400 mm in a short time, are convective and irregular, frequently causing urban flooding. These nonlinear characteristics limit traditional statistical approaches and highlight the need for AI models capable of integrating radar, satellite, and numerical forecast data to improve extreme rainfall prediction.

This study evaluates different activation functions within TDNN models for forecasting monthly maximum rainfall in Ho Chi Minh City. By comparing RMSE and R^2 across configurations, the work aims to identify the most effective activation function for extreme rainfall prediction and provide a methodological reference for future AI-based meteorological applications in Vietnam.

2. METHODS AND DATA

2.1. Data and studying area

This study employs the monthly maximum daily rainfall over a 45-year period (1980–2024), observed at the Tan Son Hoa station in Ho Chi Minh City. The location of Tan Son Hoa station is shown in Figure 1. To ensure data completeness, missing values were filled using the Expectation Maximization (EM) method [11]. Following preprocessing, the next step was to normalize the sample data into a smaller value range, specifically within $[0.1 - 0.8]$, using the Min–Max scaling technique [12], as expressed in Equation (1):

$$x' = \frac{(x - \alpha)}{(b - \alpha)} + 0.1 \quad (1)$$

Where: α is the minimum value in the dataset, b is the maximum value, x is the original value to be normalized, and x' is the normalized value.



Figure 1: Location of Tan Son Hoa meteorological station in Ho Chi Minh City

2.2. Artificial Neural Network (ANN)

Artificial Neural Networks (ANNs) are inspired by biological neural networks. They consist of neurons and connections that process information to identify the relationships between inputs and outputs. In this study, a variant of ANN is employed, called the Time Delay Neural Network (TDNN). TDNN is a special type of ANN that is particularly suitable for time series data [13,14].

The delay elements are represented by the operator **D**. Therefore, the rainfall input $R(t)$ to the i -th neuron can be determined using the following formula [15]:

$$R_i(t) = \sum_{k=0}^P w_i(k)x(t-k) + b_i \quad (2)$$

Where: $w_i(k)$ denotes the synaptic weight of the i -th neuron, and b_i represents the bias of that neuron. Subsequently, the output of the neuron U_i is obtained by processing $R_i(t)$ through a nonlinear activation function, referred to as the sigmoid function (G), as presented in Equations (3) and (4), respectively.

$$U_i = G\left(\sum_{k=0}^P w_i'(k)x(t-k) + b_i'\right) \quad (3)$$

$$G(R_i(t)) = \frac{1}{1 + e^{-R(t)}} \quad (4)$$

The output of the j -th neuron, which represents the output of the TDNN, is formulated as follows.

$$y_i(t) = F\left(\sum_{i=1}^m w_{ji}'' U_i + \alpha_j\right) \quad (5)$$

Where: $F(\cdot)$ is the activation function of the jjj -th output neuron, α_j is the bias of the output neuron, and w_{ji} is the weight between the hidden layer neurons and the output neuron.

To determine the optimal activation function for the TDNN model, multiple activation functions and network architectures were tested. Network Architecture 1 corresponds to Model C, identified as optimal in Sulaiman (2017) [11], while Network Architecture 2 is an extended version with more hidden nodes. The activation functions and configurations are summarized in Tables 1 and 2.

Model C10 consists of 12 input nodes, two hidden layers with 10 nodes each, and one output node. The inputs include maximum monthly rainfall over the previous 12 months (t to $t-11$), and the output predicts rainfall for month $t+1$. Model C15 follows the same structure but increases the hidden nodes to 15 in each hidden layer. Table 2 presents the activation functions tested for both models.

This setup contributes to advancing the application of artificial intelligence in rainfall forecasting in Vietnam and provides a practical reference for TDNN configuration in future research.

Table 1: Artificial Neural Network architecture

Model	Architecture	Training data
C10	12-10-10-1	$R_{t-11}, R_{t-10}, \dots, R_{t-2}, R_{t-1}$ and R_t
C15	12-15-15-1	$R_{t-11}, R_{t-10}, \dots, R_{t-2}, R_{t-1}$ and R_t

Table 2: List of experiments in the study

Activation function	ELU	GELU	Leakyrelu	PreLU	ReLU	RReLU	SIGMOID
Name of Test	ELU	GEL	LEA	PRE	REL	RRE	SIG

2.3. Activation functions

The activation function plays a crucial role in the training process and in improving the performance of deep artificial neural networks [16]. According to [16], the ReLU function demonstrates the best performance, whereas SELU and ELU yield comparatively lower results in experiments. In this study, several activation functions are examined within the TDNN model for rainfall prediction in Ho Chi Minh City. Detailed descriptions of these functions can be found in [16]. The list of activation functions is presented in Table 3.

Table 3: List of activation functions used in the study

No.	Function Name	Notation	Notes
1	Rectified Linear Unit	ReLU	
2	Sigmoid	Sigmoid	
3	Parametric ReLU	PreLU	The derivative function of RELU
4	Random Rectified Linear Unit	RreLU	The derivative function RELU
5	Exponential Linear Unit	ELU	
6	Leaky ReLU	LeakyRELU	The derivative function RELU
7	Gaussian Error Linear Units	GELU	

3. RESULTS AND DISCUSSION

3.1. Supplementing missing data

In this section, the study uses the EM method [11] to supplement missing data for the monthly maximum rainfall time series at Tan Son Hoa station in Ho Chi Minh City during the period 1980-2024. The missing time series is supplemented by the EM method, in which when the rainfall calculated by the EM method has a negative value, it is automatically replaced by the smallest value in the time series. The results of data supplementation are shown in Figure 2. From Figure 2, the original data is shown in the broken red line, the supplemented data is the solid blue line. The original time series has some missing months due to external factors such as forgetting to record data, data errors during the observation process, etc. After supplementing the time series continuously from January 1980 to December 2024.

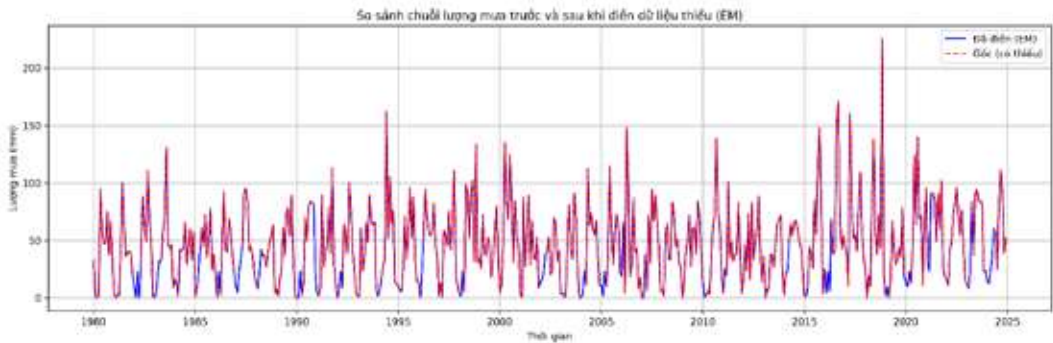


Figure 2: Monthly maximum rainfall series before and after completion of missing data

3.2. An investigation of activation functions in the TDNN model

The input data for the model is standardized according to section [12]. The TDNN model configuration used in the study includes: configuration 1 is 12-10-10-1 [11] and configuration 2 is the hidden layers are up to 15 hidden nodes. The activation functions used in the above 2 models are listed in Table 3.

With configuration 1 (C10), the training results are shown in Figure 2 and Figure 3. The results show that with this configuration, the SIG test gives results that are much different from the remaining tests, the model training process almost does not learn from the observation data, shown in the training line does not capture any peaks of the observation data (Figure 2b). Meanwhile, the remaining 6 tests have training results that match the observation data at small value ranges, not capturing the maximum peaks in the time series (Figure 2 and Figure 3). For the evaluation results using root mean square error (RMSE) and coefficient of determination (R^2 - how much of the variation (variance) of the actual data can be “captured” in the correct direction by the forecasting model), the PRE test has the best results with the smallest RMSE error among the 7 tests, and has the highest R^2 value among the 7 tests.

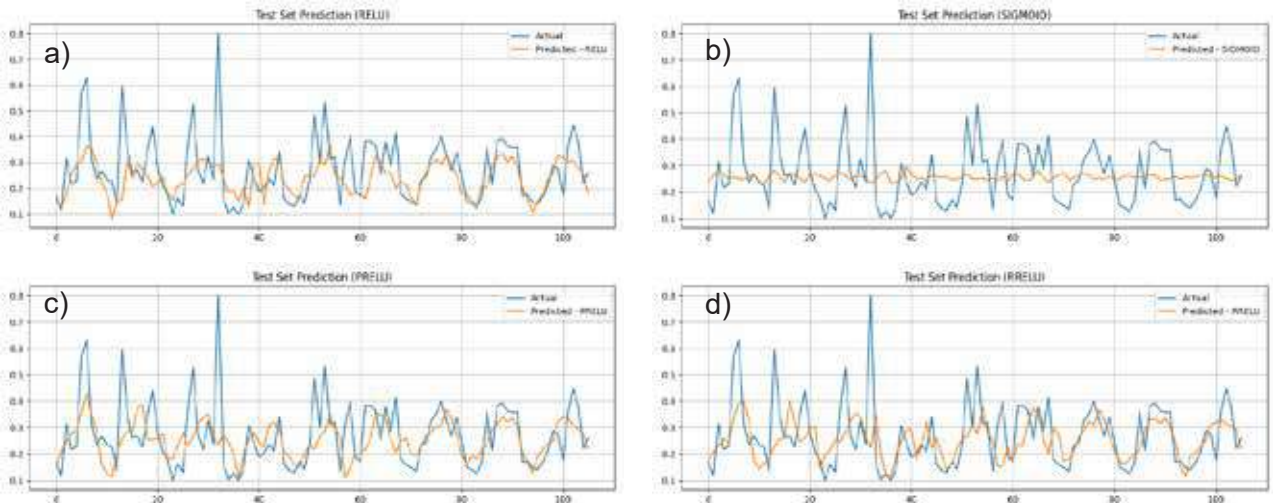


Figure 2: Training dataset results for the C10 model (a – REL; b – SIG; c – PRE; d – RRE)

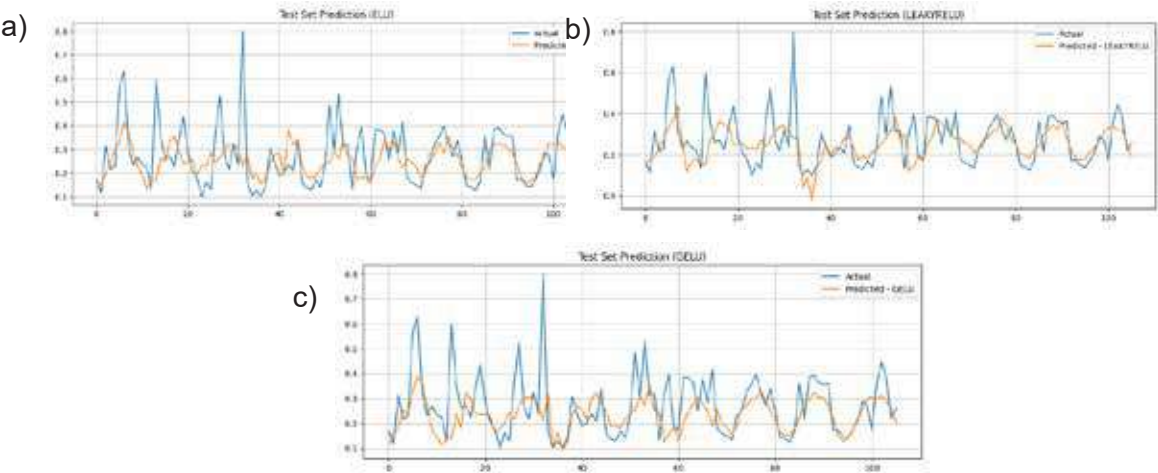


Figure 3: Training dataset results for the C10 model (a - ELU; b - LEA; c – GEL)

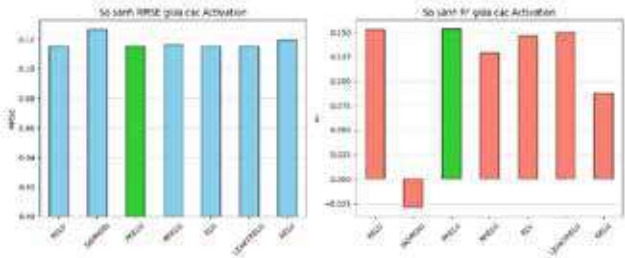


Figure 4: Comparison of RMSE and R^2 performance across activation functions in the C10 model

For configuration 2 (C15), the training results are similar to configuration C10. In configuration 2, the SIG test does not capture the same trend as the remaining tests (Figure 5b). The remaining tests capture the trend of the observation data less than 0.4, the observation values above 0.4 are not captured by all tests (Figures 5 and 6). However, when evaluated by the RMSE and R2 indexes, the RRE test gives the best results among the 7 tests with the smallest RMSE value and the highest R2 (Figure 7). When comparing the 2 configurations, the RRE test in configuration C15 gives a smaller RMSE (0.15) than the RMSE (0.18) of the PRE test in configuration C10. Similarly, the coefficient of determination value in the RRE test ($R=0.214$) of configuration 2 is larger than the R2 in the PRE test ($R=1.51$) of configuration C15.

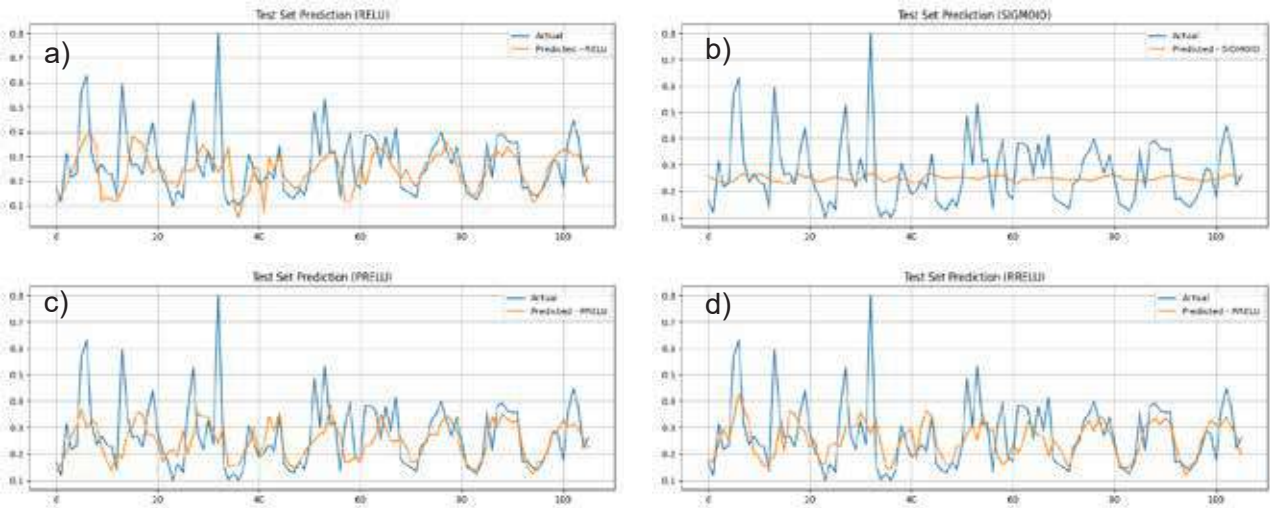


Figure 5: Training dataset results for the C15 model (a - REL; b - SIG; c - PRE; d - RRE)

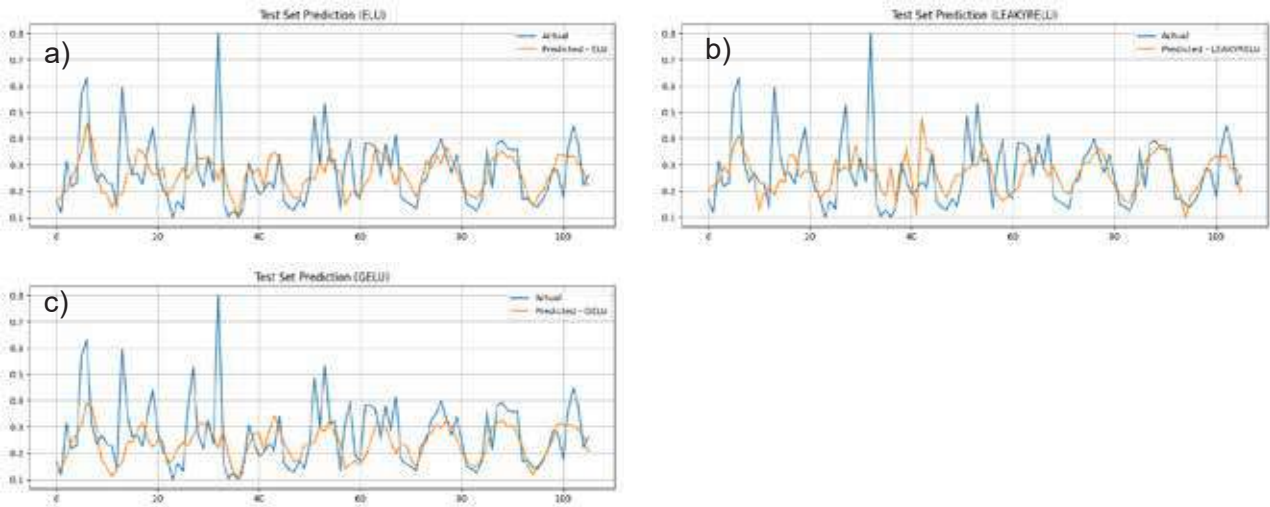


Figure 6: Training dataset results for the C15 model (a - ELU; b - LEA; c – GEL)

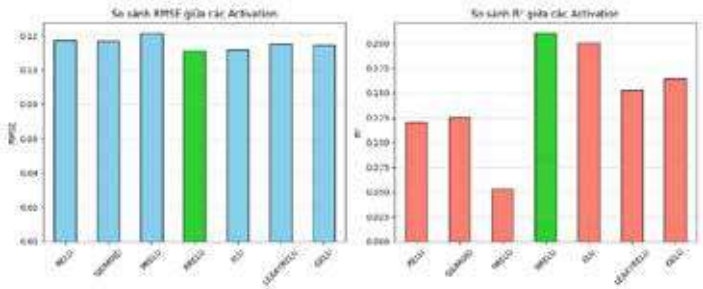


Figure 7: Comparison of RMSE and R² performance across activation functions in the C15 model

4. CONCLUSION

This study evaluated seven activation functions in the TDNN model across two configurations for forecasting maximum monthly rainfall in Ho Chi Minh City. The results indicate that configuration 2 (1-15-15-1) with the RRELU activation function achieved the lowest RMSE and the highest R^2 among all tested models. These outcomes provide a useful reference for future rainfall forecasting research in the area. However, due to limited data coverage, the coefficient of determination remains low ($R^2 = 0.214$), suggesting that the RRELU-based TDNN model has not yet captured rainfall variability effectively. Future work will focus on expanding the dataset and improving model performance to enhance forecast accuracy.

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